BTVNANOCOMMISSIONING · ZB_ZC_RATIO WORKFLOW

Z(→ℓℓ) + b/c jets in Run 3: first look at data

Summer22

2022 C–D data vs MC, a small sample (0.7 fb^{-1} in Z→μμ, 0.4 fb^{-1} in Z→ee), AK4 jets

26 October 2026

What is new since last time

DATA

2022 C–D

First look at data. Muon and EGamma, 22Sep2023 reprocessing.

CAMPAIGN

Summer22

One of four campaigns run (Summer22, Summer22EE, Summer23, Summer23BPix). Each has its own corrections and its own deck; they are not combined.

CORRECTIONS

Full set

Jet energy corrections, jet veto maps, lepton energy scale, lepton and trigger SFs, all from the official POG files. Hsin-Wei's latest code merged in.

This is a small, quick run: 20 files per MC sample and 20 data files per channel and campaign, run interactively. The aim is to see whether data and MC agree before we scale up. The selection is the same as in the MC-only study: two leptons on the Z peak and at least one AK4 jet with pT > 30 GeV. No b- or c-tag requirement yet.

Data used

Dataset (DAS)	Files	Lumisections	brilcalc [pb⁻¹]	Factor	Used [pb⁻¹]
/EGamma/Run2022C-22Sep2023-v1/NANOAOD	15	1,389	224	1	224
/EGamma/Run2022D-22Sep2023-v1/NANOAOD	5	620	165	1	165
/Muon/Run2022C-22Sep2023-v1/NANOAOD	12	1,861	431	1	431
/Muon/Run2022D-22Sep2023-v1/NANOAOD	8	1,102	311	1	311

Summer22: Z→μμ 742 pb⁻¹, Z→ee 389 pb⁻¹

The 22Sep2023 re-reconstruction in NanoAODv12, 20 files per channel spread over the eras. Z→μμ comes from the Muon PD (IsoMu24), Z→ee from the EGamma PD (Ele30_WPTight). Muon and EGamma files cover different lumisections, so each channel has its own luminosity and the MC is scaled to it.

Golden JSON and luminosity

GOLDEN JSON

Cert_Collisions2022_355100_362760_Golden.json

The official 2022 golden JSON, read from the CMS central area on cvmfs (cms-griddata …/DC/Collisions22/latest). This is the newest version, after the 2026 tracker re-review.

LUMINOSITY

brilcalc, normtag_PHYSICS

Run in the BRIL container on LPC, on exactly the lumisections we processed. Muon and EGamma get their own numbers.

Checked against the newest golden JSON

All 4,828 processed lumisections are golden in the newest version.

MC samples

Sample	Generator	Run3Summer22NanoAODv12-…	σ [pb]	Files
DYto2L-2Jets_MLL-50	amcatnloFXFX-pythia8	130X_mcRun3_2022_realistic_v5-v5	6346	20 / 525
DYto2L-2Jets_MLL-50_PTLL-100	amcatnloFXFX-pythia8	130X_mcRun3_2022_realistic_v5-v3	105	20 / 418
DYto2L-2Jets_MLL-50_PTLL-200	amcatnloFXFX-pythia8	130X_mcRun3_2022_realistic_v5-v2	11.24	20 / 335
DYto2L-2Jets_MLL-50_PTLL-400	amcatnloFXFX-pythia8	130X_mcRun3_2022_realistic_v5-v2	0.6433	20 / 424
DYto2L-2Jets_MLL-50_PTLL-600	amcatnloFXFX-pythia8	130X_mcRun3_2022_realistic_v5-v2	0.09941	20 / 185
TTto2L2Nu	powheg-pythia8	130X_mcRun3_2022_realistic_v5-v2	98.88	20 / 67
WW	pythia8	130X_mcRun3_2022_realistic_v5-v2	75.9	not run
WZ	pythia8	130X_mcRun3_2022_realistic_v5-v2	27.6	20 / 45
ZZ	pythia8	130X_mcRun3_2022_realistic_v5-v2	12.17	20 / 21

Full DAS name: /<sample>_TuneCP5_13p6TeV_<generator>/Run3Summer22NanoAODv12-<processing>/NANOAODSIM. Files: used / available. The DY pT($\ell\ell$) samples are stitched: each binned sample keeps only its own LHE pT($\ell\ell$) slice and the inclusive one covers pT($\ell\ell$) < 100 GeV. DY events are split by the hadron flavour of the leading jet (b, c or light). $t\bar{t}$ and dibosons are not split. Cross sections as in the MC-only study. WW not run here (too slow to read from remote sites).

Corrections and scale factors

Correction	Applied to	POG file	Key / tag
Jet energy corrections	data + MC	JME jet_jerc	Summer22_22Sep2023_V4 (data: with residuals)
Jet energy resolution	MC	JME jet_jerc, jer_smear	Summer22_22Sep2023_JRV2
MET	data + MC	from the corrected jets	type-1 PuppiMET
Jet veto map	data + MC	JME jetvetomaps	Summer22_23Sep2023_RunCD_V1
Muon momentum scale and resolution	data + MC	MUO muon_scalesmearing	Run 3 recipe
Electron energy scale / smearing	data / MC	EGM electronSS_EtDependent	Scale / SmearAndSyst
Pileup weight	MC	LUM puWeights	Collisions2022_355100_357900_eraBCD_GoldenJson
Muon ID and isolation SF	MC	MUO muon_Z	NUM_TightID_DEN_TrackerMuons, NUM_TightPFIso_DEN_TightID
Muon trigger SF (leading muon)	MC	MUO muon_Z	NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
Electron ID and reco SF	MC	EGM electron	Electron-ID-SF: Tight and Reco (2022Re-recoBCD)
Electron trigger SF (leading electron)	MC	EGM electronHlt	HLT_SF_Ele30_TightID
b- and c-tag SF	MC	BTV btagging, ctagging	not applied yet (no tag cut)

Every file is the official one on cvmfs: /cvmfs/cms-griddata.cern.ch/cat/metadata/<POG>/Run3-22CDSep23-Summer22-NanoAODv12/latest/. The framework uses them for all BTV workflows.

POG recommendations vs what we do (1/2)

Data and jets

Topic	POG recommendation	What we do	Status
Golden JSON (DC)	Use the latest golden JSON	Latest version from cvmfs	**Same**
Luminosity (LUM)	brilcalc with normtag_PHYSICS	Same, on the lumisections we processed	**Same**
Pileup (LUM)	Official pileup weights	Same file	**Same**
Jet energy corrections (JME)	Summer22_22Sep2023_V4, resolution Summer22_22Sep2023_JRV2	Same tags	**Same**
Jet ID (JME)	Tight lepton veto ID; the NanoAODv12 jetId branch needs the JME fix	Fix applied	**Same**
Jet veto map (JME)	Veto the event if a jet with pT > 15 GeV, tight ID, EM fraction < 0.9 and away from muons falls in the map	Framework vetoes on any jet in the map. Drops about 6% of Z→μμ events instead of 1–2%; data/MC moves by 0.3% or less	**Differs**
Rare high-pT jets (JME)	Reject jets with rawFactor > 0.9 (known JEC issue)	Not applied. About 1 event in a million, mostly forward, outside $\|\eta\| < 2.5$	**Not yet**

Read on 26 October 2026: BTV scale factor page, MUO usage recommendations, JME jet energy scale page, EGamma Run 3 TWiki (last edited 2023, older than the cvmfs files we use). Jet veto recipe from the JME jet veto map recommendation. The 6% and 1–2% are quick estimates on one chunk.

POG recommendations vs what we do (2/2)

Leptons and tagging

Topic	POG recommendation	What we do	Status
Muon ID and isolation (MUO)	Tight ID, PF isolation < 0.15 (Tight), SFs from muon_Z	As recommended	**Same**
Muon trigger SF (MUO)	IsoMu24 SF per muon (Tight ID + Tight iso)	Leading muon only (pT > 35 GeV, on the plateau). Effect of using both muons not checked yet	**Simplified**
Electron ID and reco (EGM)	Cut-based Tight ID, SFs for the 22Sep2023 re-reco	As recommended	**Same**
Electron trigger SF (EGM)	HLT_SF_Ele30_TightID per electron	Leading electron only. Using both electrons raises the Z→ee MC by 1.1–1.9% (quick estimate)	**Simplified**
Lepton energy scale (EGM, MUO)	Scale in data, smearing in MC	Same files and recipe. The Z→ee peak is 1.5–4% wider in MC than in data (all four campaigns), to be understood	**Check**
b/c tagging (BTV)	AK4 PUPPI, pT > 20, |η| < 2.5, tight ID, hadronFlavour. PNet b: fixed-WP and shape SFs; PNet c: fixed-WP SFs only	pT > 30, |η| < 2.5, tight ID, hadronFlavour. No tag SFs yet (no tag cut)	**Not yet**
L1 prefiring	No prefiring weight in Run 3	None	**Same**

c-tagging has no shape SFs for 2022/2023, only fixed working points. A template fit in the CvL–CvB plane needs a plan for that. UParT SFs exist only for NanoAODv15.

What Hsin-Wei's update brought

We merged her latest coffea_machine branch (5 commits, up to 1 October) into ours on 26 October, before this run.

FIXED FOR EVERYONE

Electron SFs with two electrons

The shared electron SF code used to keep only the last electron's SF. Her fix multiplies the SFs of all electrons. We had worked around it by calling it once per electron, so our numbers do not change, but the common code is now right.

SAFER

Empty lepton slots

The SF code now fills empty lepton slots before calling the POG files. The 2023 reco SFs also need the electron φ, and that path is now protected too.

TO REUSE

Matched electron trigger SF

Her Run 2 electron trigger SF takes the electron that fired the trigger, matched to the HLT object. That is the right way to do it, and we plan to bring it to Run 3 instead of using the leading electron.

Bottom line

Her Run 2 electron SF files and settings do not touch the Run 3 inputs, and both branches now share one correction code. Where files clashed we kept our validated Run 2 selection. One thing to tell her: in the muon SF code the trigger-SF pT clip gets overwritten (we do not use that path).

Cutflow: data vs MC

	trigger	leptons	OS	Zmass	MET	jet
Z→ee data	5,081,424	115,201	114,206	108,925	105,281	18,384
MC	410,352	127,664	126,896	121,788	113,419	21,473
data / MC	12.38	0.90	0.90	0.89	0.93	0.86
Z→μμ data	8,958,823	401,972	401,814	380,702	369,980	63,056
MC	994,520	420,139	420,087	403,664	377,307	69,404
data / MC	9.01	0.96	0.96	0.94	0.98	0.91

Data/MC only means something from the two-lepton step on. At the trigger step the data are mostly single-lepton events (W→ℓν, multijet) that fire the same trigger but are not simulated: we only have DY, $t\bar{t}$ and VV. The two-lepton cut keeps 2–5% of the triggered data but 30–40% of the MC, and data/MC drops from about 10 to about 1. MC here is before SFs (generator weight × σ × L / Σw only), so data/MC is expected a few % off 1 at the lepton steps.

Yields after the full selection

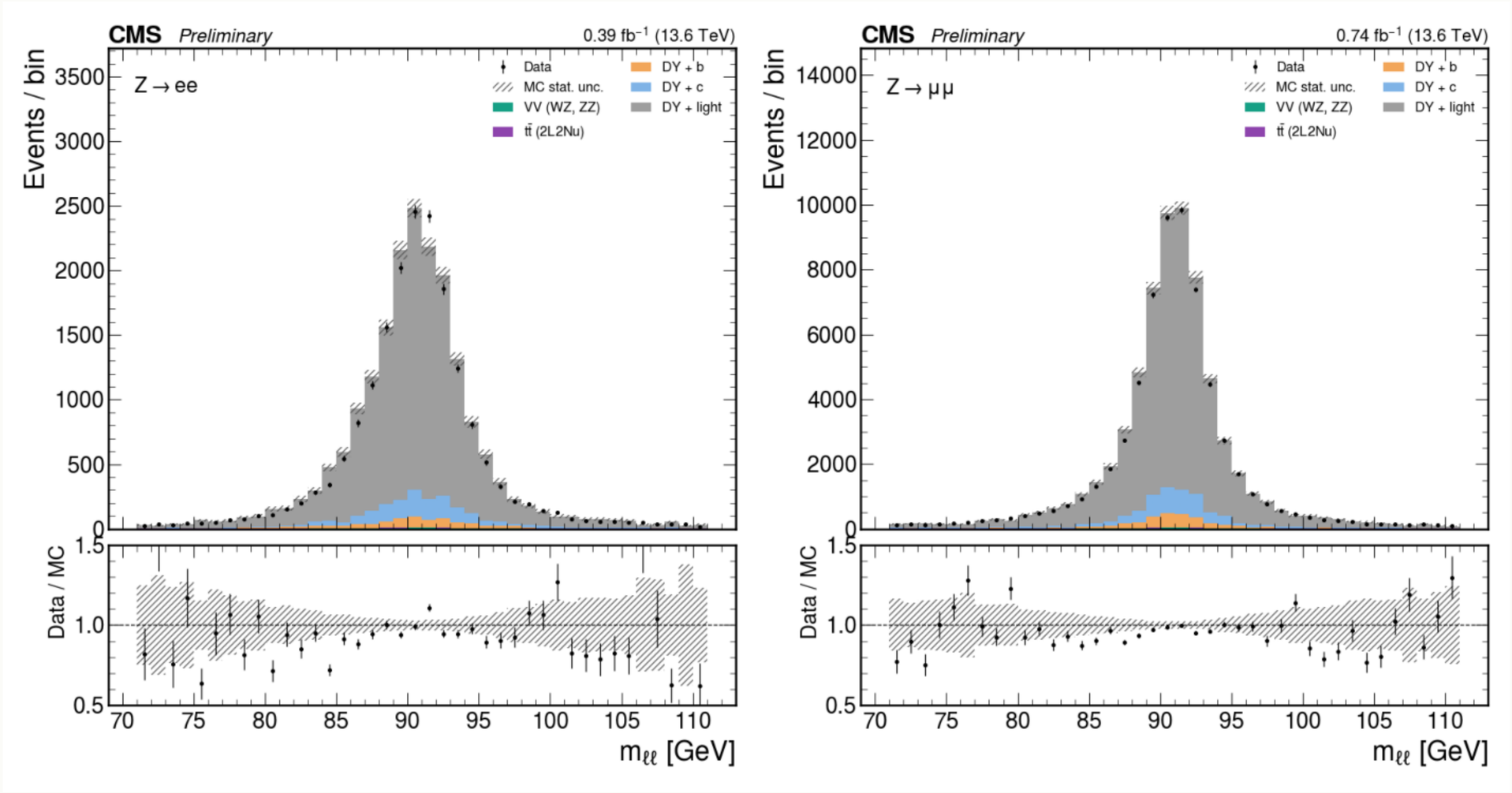
	DY + light	DY + c	DY + b	tt̄	VV	MC	Data	Data/MC
Z→ee	16,754	1,530	710	71	59	19,122	18,384	0.961
Z→μμ	56,938	5,304	2,760	246	202	65,449	63,056	0.963

Z→ee: DY + b is 3.7% and DY + c is 8.1% of DY (c/b = 2.16); tt̄ + VV is 0.7% of the MC.

Z→μμ: DY + b is 4.2% and DY + c is 8.2% of DY (c/b = 1.92); tt̄ + VV is 0.7% of the MC.

Dilepton mass

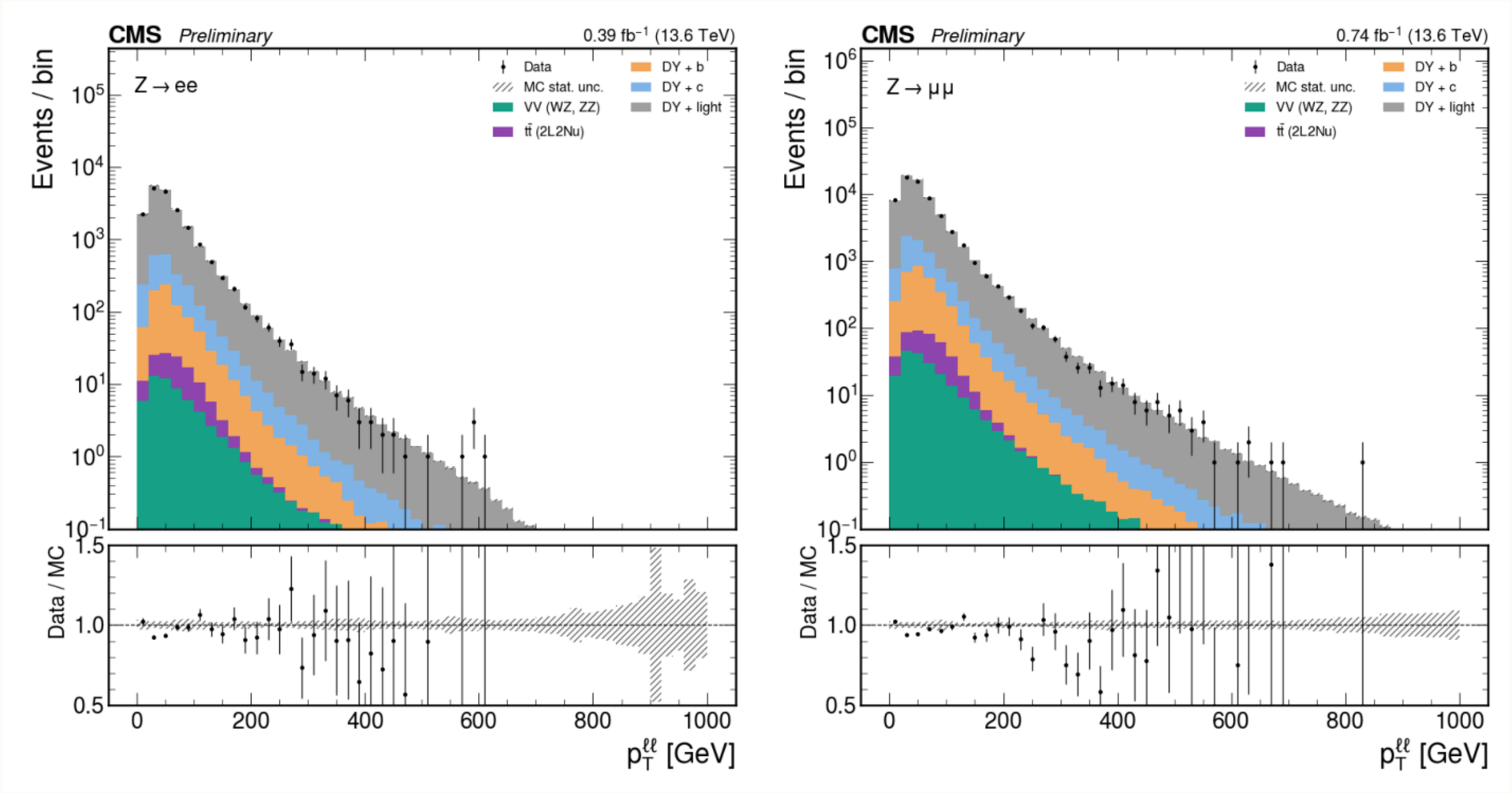
The Z peak in data and MC. Linear y-axis.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 12

Dilepton pT

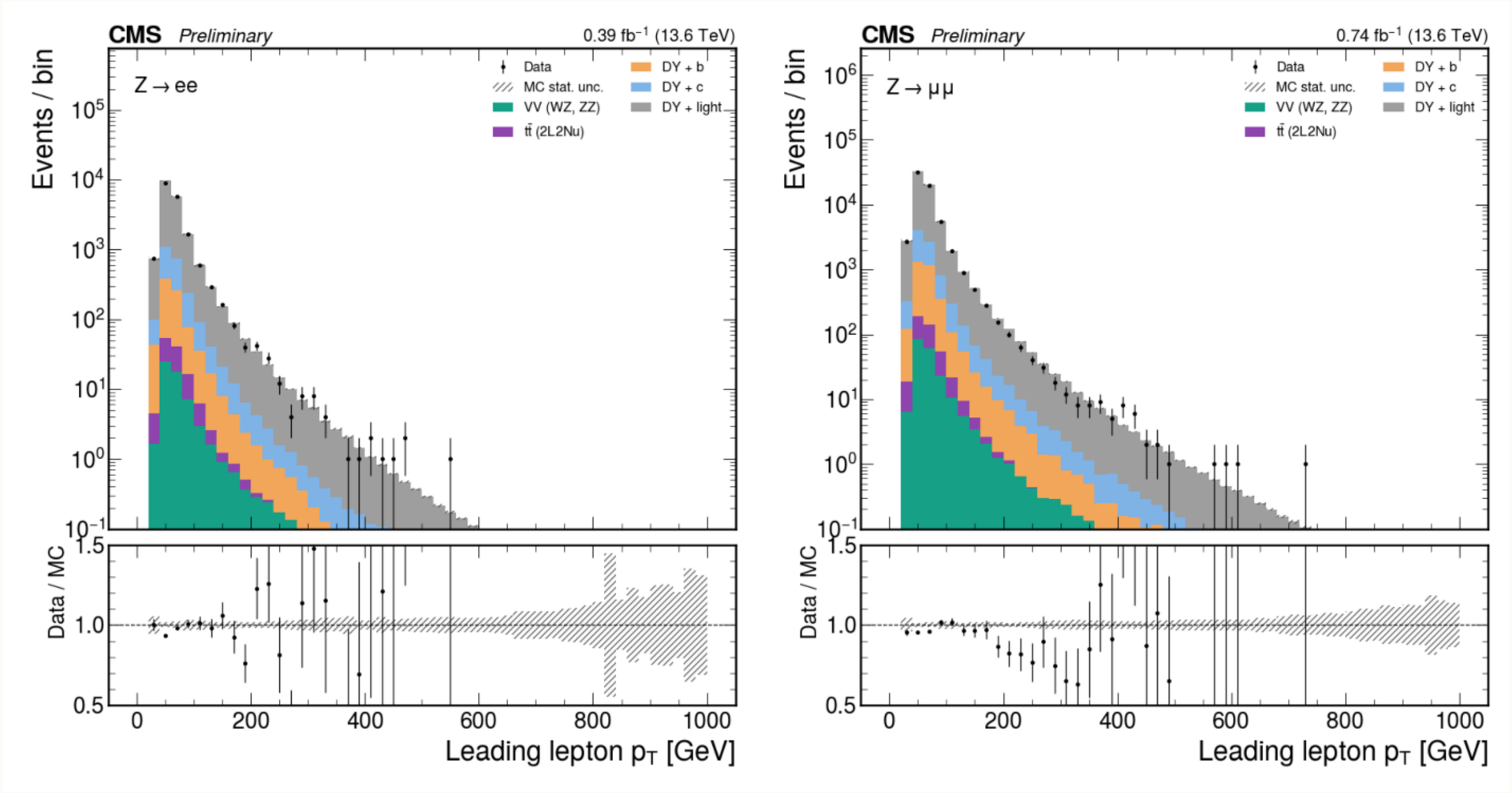
pT of the Z candidate. No cut on it.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 13

Leading-lepton pT

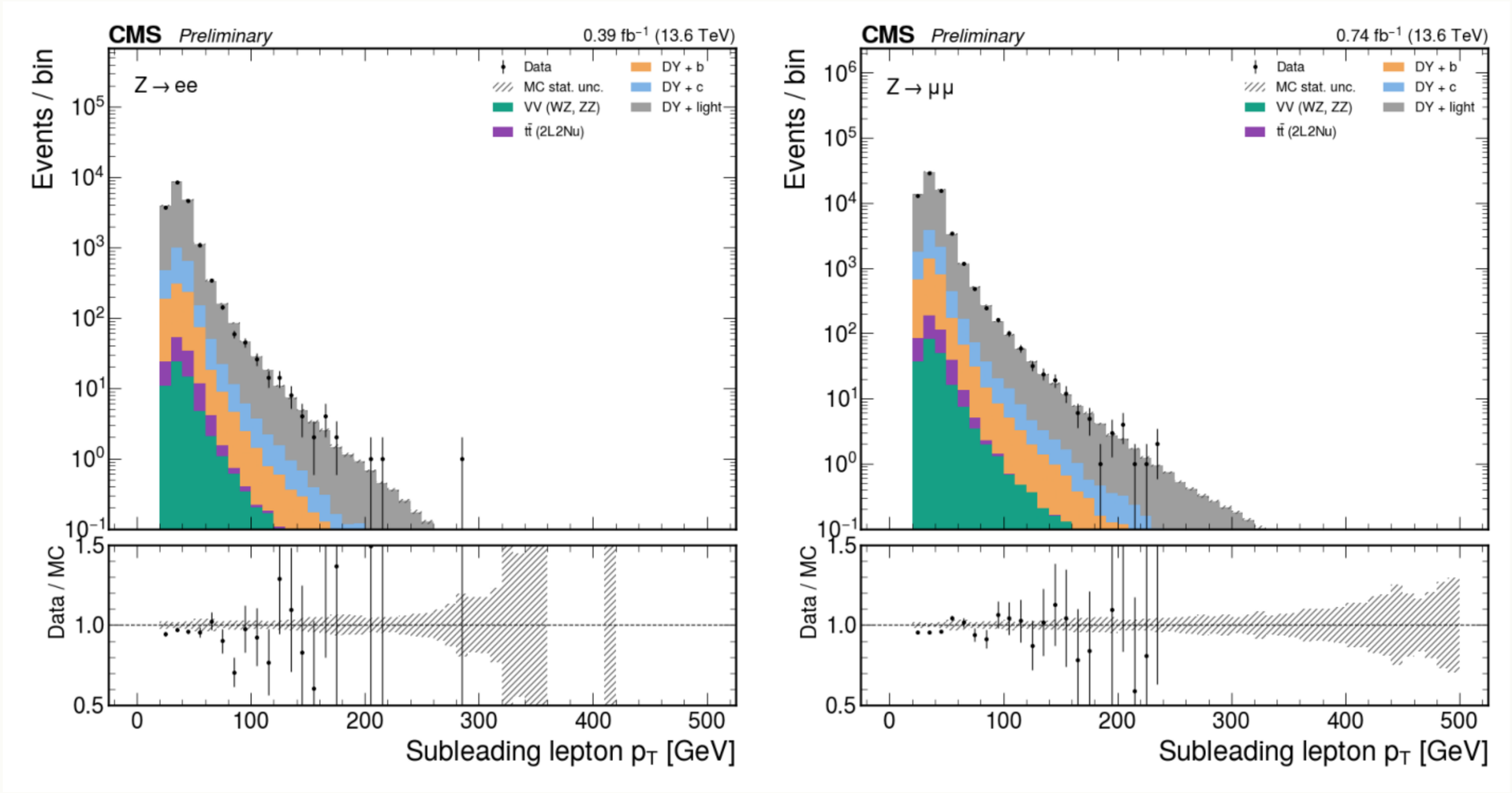
Leading lepton, pT > 35 GeV required.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 14

Subleading-lepton pT

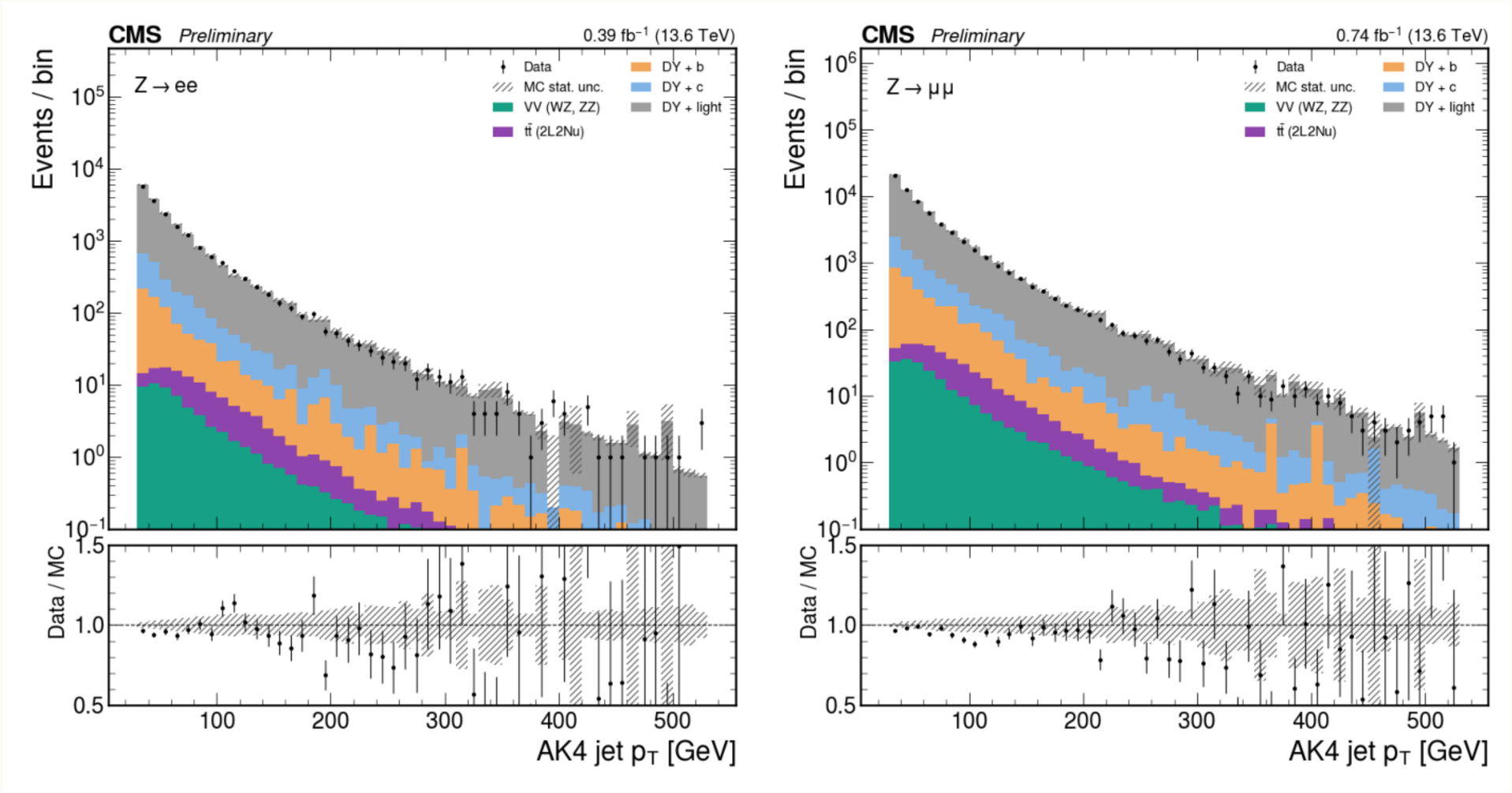
Subleading lepton, pT > 25 GeV required.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 15

Leading AK4 jet pT

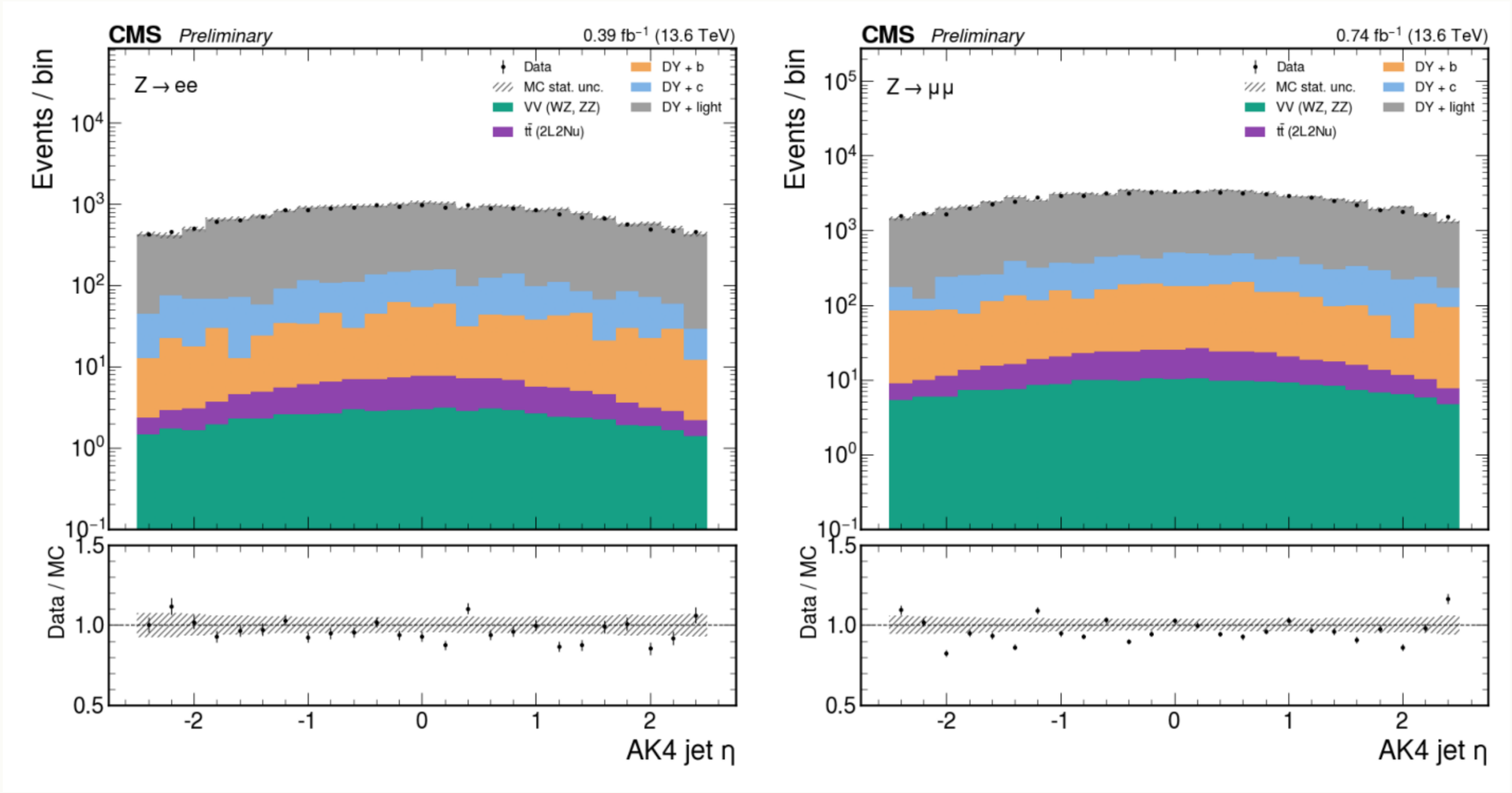
Leading lepton-cleaned AK4 jet, pT > 30 GeV required.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 16

Leading AK4 jet η

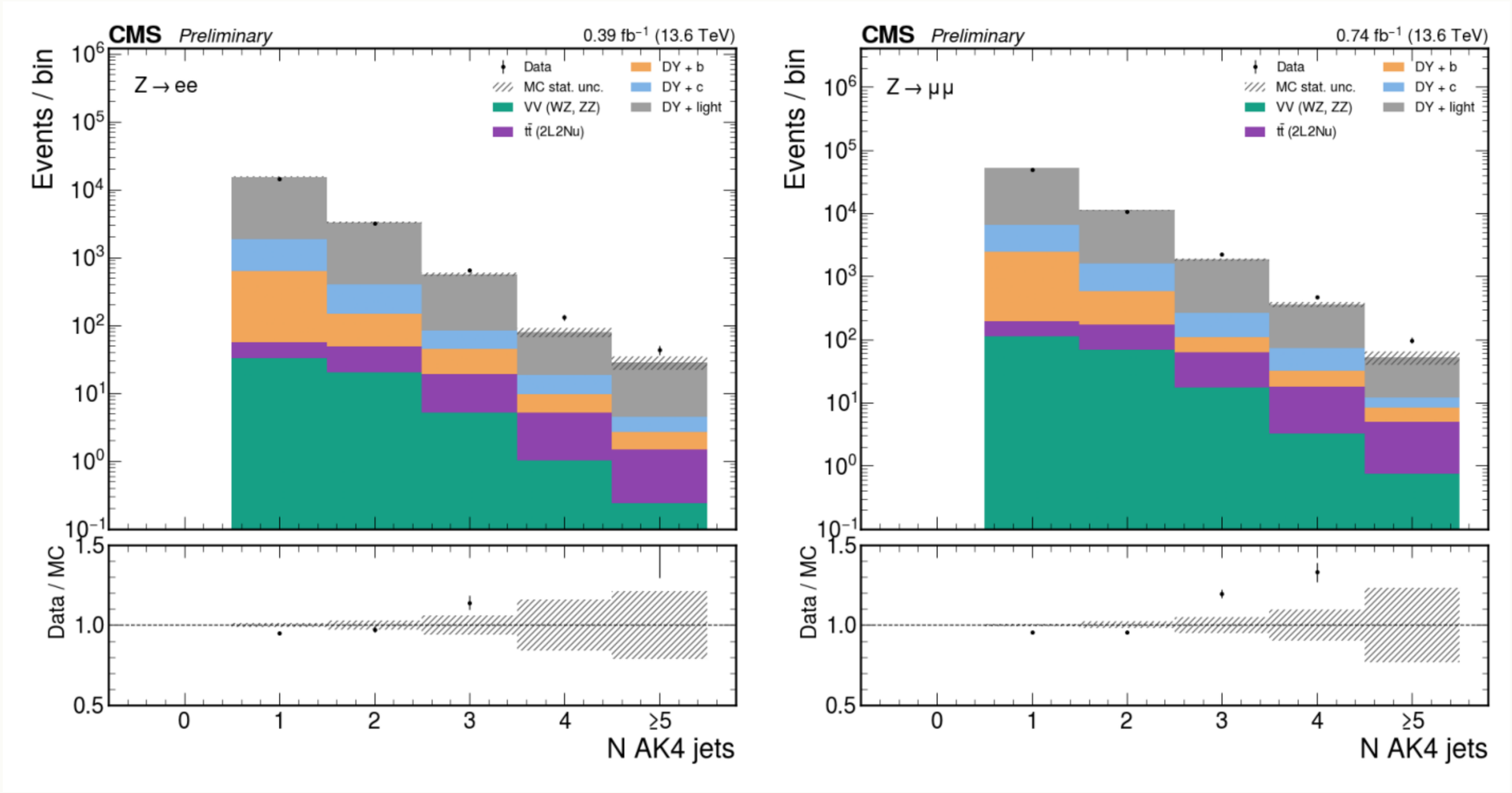
Pseudorapidity of the leading jet, |η| < 2.5.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 17

Number of AK4 jets

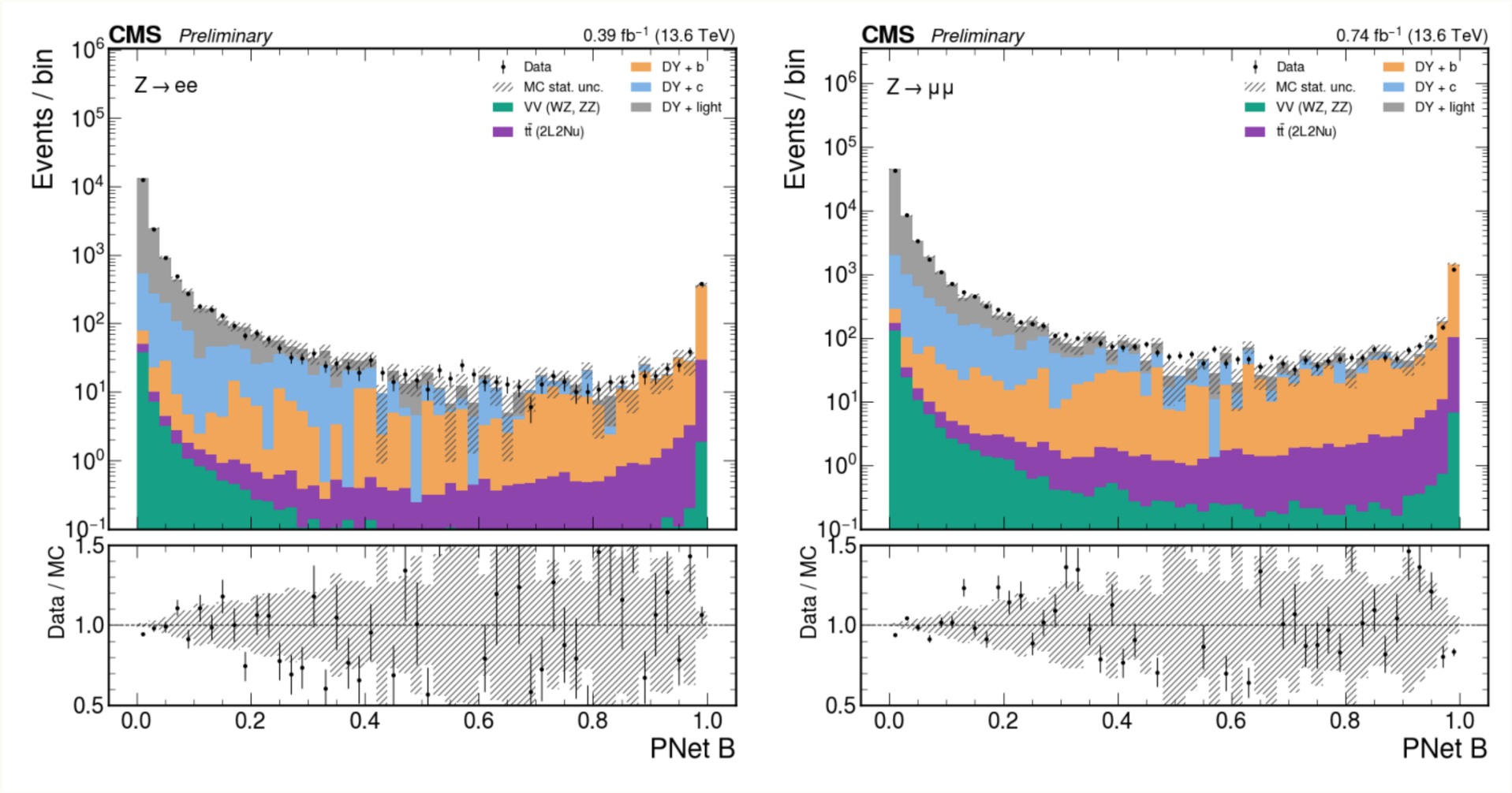
Jets with pT > 30 GeV and |η| < 2.5. The last bin holds 5 or more jets.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 18

PNet b-tag score

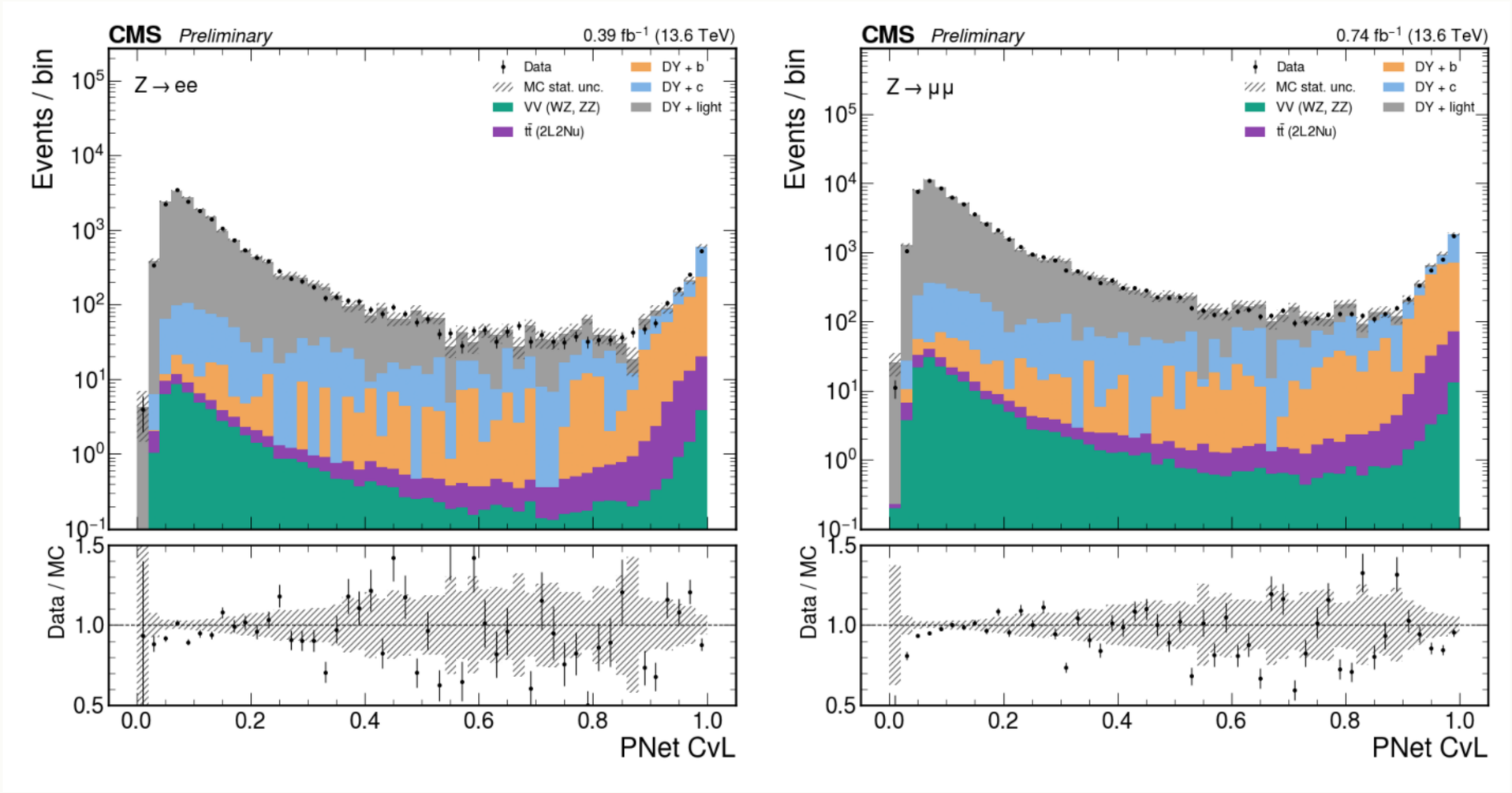
Leading jet. b jets sit near 1, light jets near 0.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 19

PNet charm vs light

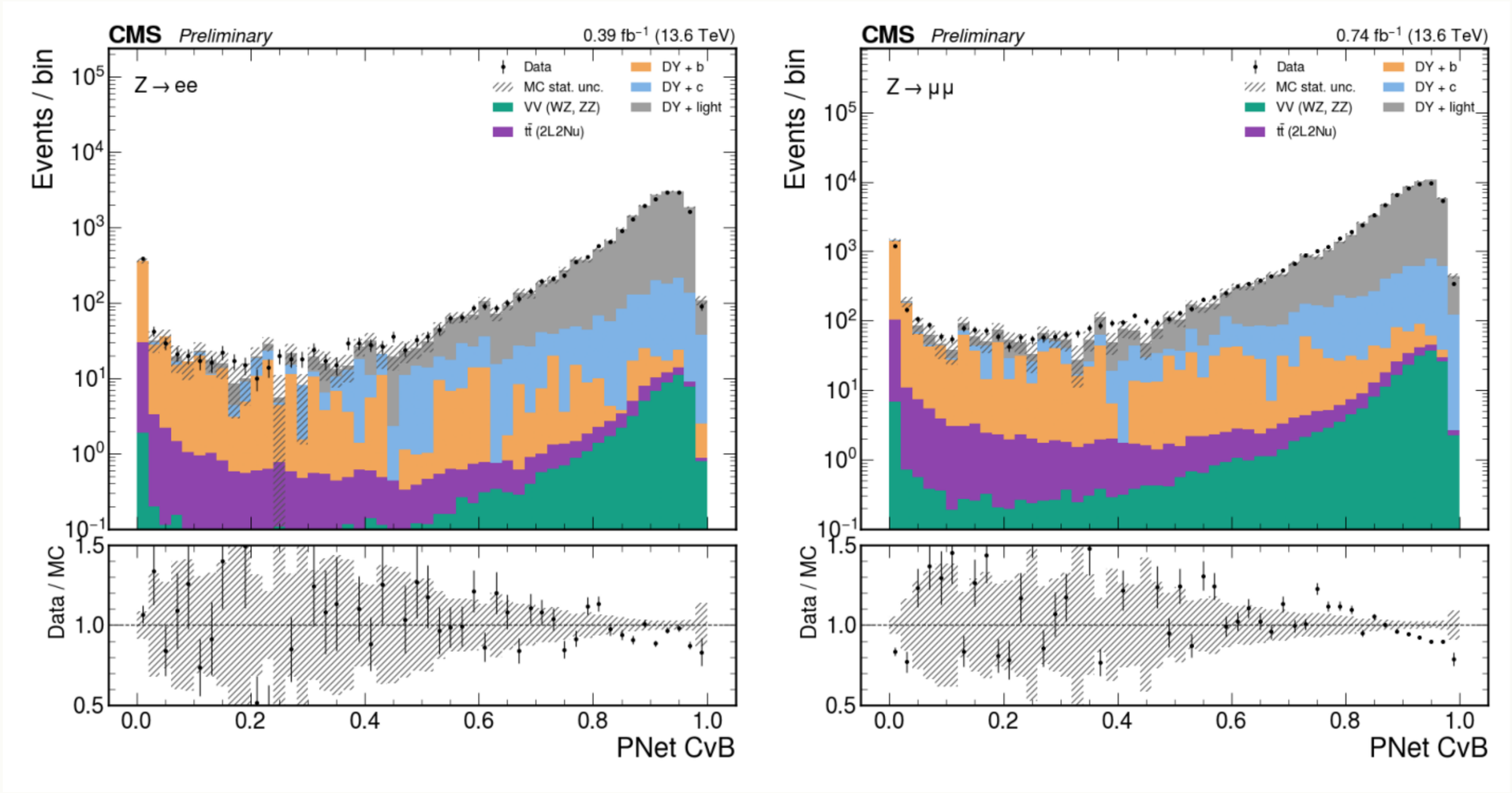
Leading jet.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 20

PNet charm vs b

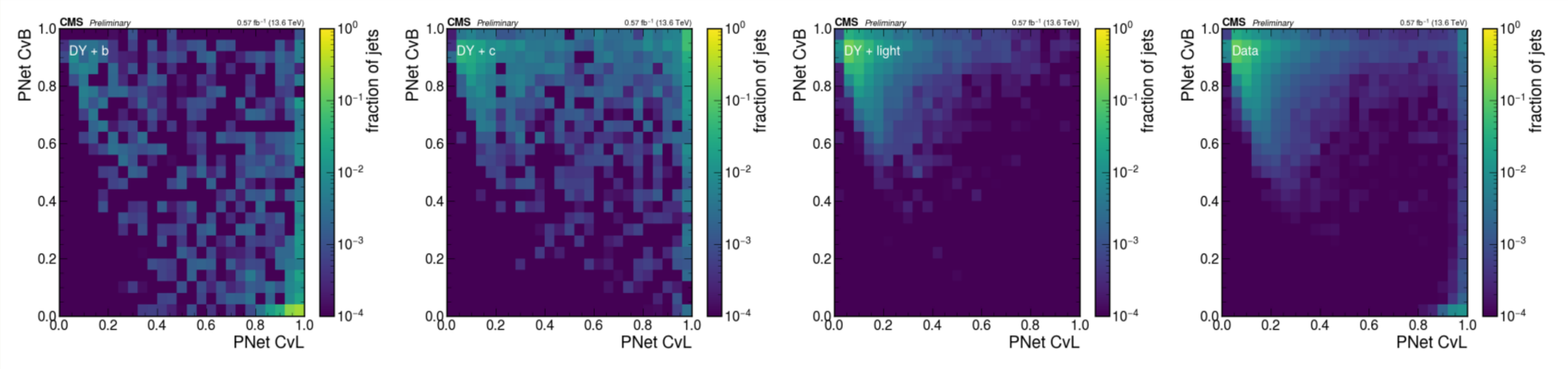
Leading jet.



Run 3, 13.6 TeV · 2022 C–D · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 21

PNet charm-tagging plane: CvL vs CvB

Leading jets, both channels summed, each panel normalized to unit area. Data is a mix of all flavours, mostly light.



Summary and next steps

Data and MC agree to about 4% in both channels

With 0.39 fb^{-1} in Z→ee and 0.74 fb^{-1} in Z→μμ, data/MC after the full selection is 0.96 ± 0.01 in both channels. The Z peak, the jet pT and η and the PNet scores have the same shape in data and MC. DY + b is about 4% and DY + c about 8% of the DY events. The 4% deficit is more than the statistical uncertainty, but of the size the DY cross section, luminosity and lepton SFs allow together. We are on track to scale up.

- Full run on condor, after fixing the golden JSON path and moving to the JME-style jet veto.
- Electron trigger SF from both electrons, matched to the trigger object as in Hsin-Wei's Run 2 code.
- Find out why the Z→ee peak is wider in MC than in data (electron smearing).
- WW was not run here (too slow to read); add it in the full run.
- b/c-tag SFs once we cut on the tagger; decide what to do without c-tag shape SFs.